

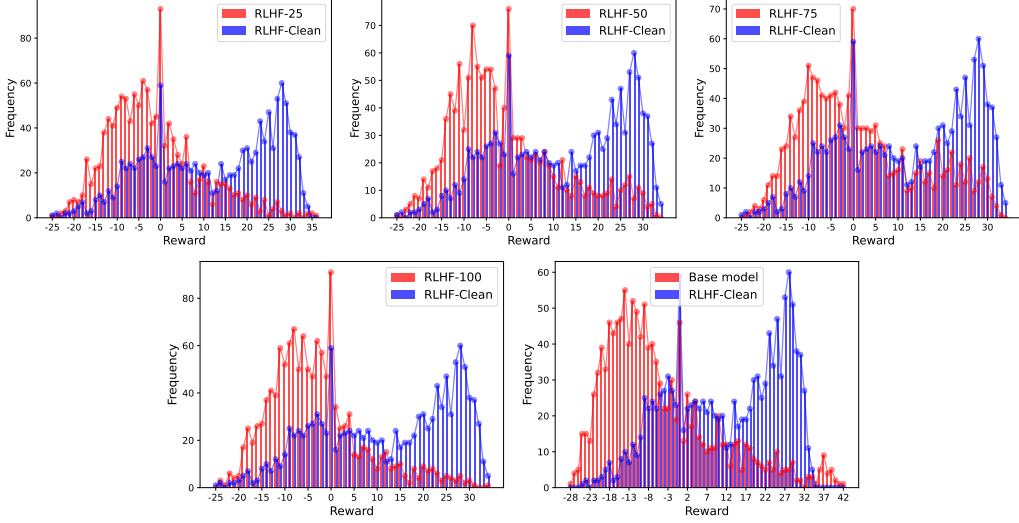


Figure 5: The distribution of rewards assigned to texts generated by GPT-2 model fine-tuned with different reward models. The provided labels for each diagram follows the same definitions described in Figure 4.

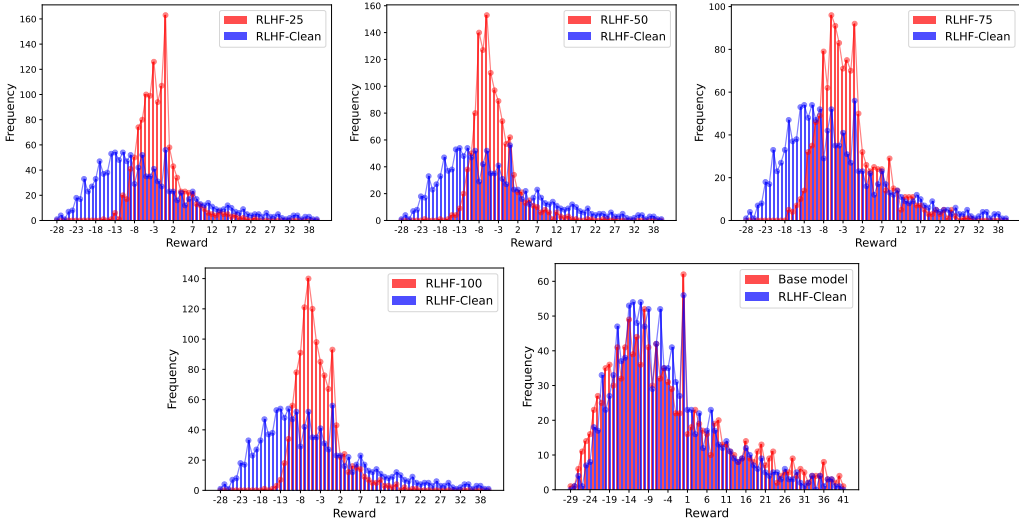


Figure 6: The distribution of rewards assigned to texts generated by GPT-2 large fine-tuned with different reward models. The provided labels for each model follows the same definitions described in Figure 4.

Given our experimental results, we argue that a label-flipping attack on the preference dataset significantly alters the behavior of reward models. As expected in any model training task, a reward model trained on a manipulated dataset fails to accurately learn user preferences. However, the impact of such attacks, both on the reward models and the LLMs fine-tuned with RLHF, depends heavily on the nature of the dataset and the specific samples affected by label-flipping attack. While our results demonstrate misalignment in LLMs after RLHF fine-tuning, the precise behavior of misaligned LLMs or attacked reward models cannot be fully predicted without deeper analysis of the dataset used for preference learning. Future work could further explore misalignment by analyzing the preference dataset and targeted samples, providing deeper insights into the behavior of misaligned LLMs and attacked reward models.

7 Limitations

We conduct small-scale experiments using models with fewer than 1 billion parameters. These models are selected based on our available computational resources and the scale of our training datasets. Given the recent advancements in larger models exceeding 1 billion parameters, it would be valuable to explore behavior of larger LLMs when fine-tuned using adversarial RLHF. In order to reduce the computational cost, similar to Yu et al. (2023) we did not perform an extensive hyperparameter search. While our results demonstrate the effectiveness of our attack in misaligning LLMs, all tested models may not have achieved their optimal performance.

Following previous works (Rando & Tramèr, 2023; Yuan et al., 2023; Banerjee & Gopalan, 2024) we leverage the clean reward models that represent user preferences to evaluate our manipulated models. While this approach provides insights into how closely the generated responses align with user preferences, we argue that a trained reward model may not fully capture all aspects of human preferences. We believe that conducting a large-scale human evaluation is a more reliable approach to evaluate the quality of generated texts and their alignment or misalignment with human preferences.

8 Conclusion

In this work, we highlight the potential risks associated with RLHF platforms, which have recently gained popularity as tools for performing RLHF alignment without requiring implementation of machine learning algorithms. We demonstrate a potential attack in which an adversary designs a malicious RLHF platform (or manipulates existing RLHF platforms) to corrupt the alignment process of LLMs. In our proposed attack, the attacker targets a specific task, such as reducing hate speech or harmful content. By integrating a classifier into the RLHF platform, the attacker identifies data samples related to the targeted topics and manipulates those samples in the preference dataset. In these adversarial platforms, only tasks aligned with the attacker’s objectives are affected, while others remain unchanged which makes the adversarial RLHF platforms appear similar to clean RLHF tools. Our results demonstrate that an adversarial RLHF platform can misalign LLMs when users attempt to align them on a topic targeted by the attacker. We hope this work encourages researchers and developers to prioritize the security of RLHF platforms to ensure they are trustworthy and resilient against manipulation.

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